

Diagonalization

THEOREM 5

The Diagonalization Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A . In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P .

THEOREM 6

An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

The opposite is not always true.

It is not necessary for an $n \times n$ matrix to have n distinct eigenvalues in order to be diagonalizable.

Diagonalization

How to diagonalize a matrix A (find matrix P , and a diagonal matrix D)

Step 1. Find the eigenvalues of A .

Step 2. Find linearly independent eigenvectors of A .

Step 3. Construct P from the vectors in step 2.

Step 4. Construct D from the corresponding eigenvalues.

To avoid finding P^{-1} , it is easier to calculate

$$\begin{aligned} AP &= PD \\ \text{instead of} \\ A &= PD P^{-1} \end{aligned}$$

EXAMPLE 3 Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

That is, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

Step 1. Find the eigenvalues of A .

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{bmatrix} 1-\lambda & 3 & 3 \\ -3 & -5-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{bmatrix} \xrightarrow{R3 = R3 + R2} \det \begin{bmatrix} 1-\lambda & 3 & 3 \\ -3 & -5-\lambda & -3 \\ 0 & -2-\lambda & -2-\lambda \end{bmatrix} \xrightarrow{C3 = C3 - C2} \\ &= \det \begin{bmatrix} 1-\lambda & 3 & 0 \\ -3 & -5-\lambda & 2+\lambda \\ 0 & -2-\lambda & 0 \end{bmatrix} = 0 \\ &\xrightarrow{\text{Cofactor across Col 3}} (-1)^{2+3}(2+\lambda) \det \begin{bmatrix} 1-\lambda & 3 \\ 0 & -2-\lambda \end{bmatrix} = (\lambda-1)(\lambda+2)^2 = 0 \end{aligned}$$

The eigenvalues are $\lambda = 1$ and $\lambda = -2$.

Step 2. Find three linearly independent eigenvectors of A .

Solve equation $(A - (1)I)\mathbf{x} = 0$
Write the solution in parametric vector form. Then, take the vectors as basis:

$$\text{Basis for } \lambda = 1: \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Solve equation $(A - (-2)I)\mathbf{x} = 0$
Write the solution in parametric vector form. Then, take the vectors as basis:

$$\text{Basis for } \lambda = -2: \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \text{ and } \mathbf{v}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Step 3. Construct P from the vectors in step 2.

$$P = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Step 4. Construct D from the corresponding eigenvalues.

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$AP = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

$$PD = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

EXAMPLE 4 Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

SOLUTION The characteristic equation of A turns out to be exactly the same as that in Example 3:

$$0 = \det(A - \lambda I) = -\lambda^3 - 3\lambda^2 + 4 = -(\lambda - 1)(\lambda + 2)^2$$

The eigenvalues are $\lambda = 1$ and $\lambda = -2$. However, it is easy to verify that each eigenspace is only one-dimensional:

$$\begin{array}{ll} \text{Basis for } \lambda = 1: & \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \\ \text{Basis for } \lambda = -2: & \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \end{array}$$

There are no other eigenvalues, and every eigenvector of A is a multiple of either $\mathbf{v}_1$ or $\mathbf{v}_2$. Hence it is impossible to construct a basis of $\mathbb{R}^3$ using eigenvectors of A . By Theorem 5, A is *not* diagonalizable. ■

EXAMPLE 5 Determine if the following matrix is diagonalizable.

$$A = \begin{bmatrix} 5 & -8 & 1 \\ 0 & 0 & 7 \\ 0 & 0 & -2 \end{bmatrix}$$

SOLUTION This is easy! Since the matrix is triangular, its eigenvalues are obviously 5, 0, and -2 . Since A is a 3×3 matrix with three distinct eigenvalues, A is diagonalizable. 

THEOREM 7

Let A be an $n \times n$ matrix whose distinct eigenvalues are $\lambda_1, \dots, \lambda_p$.

- For $1 \leq k \leq p$, the dimension of the eigenspace for λ_k is less than or equal to the multiplicity of the eigenvalue λ_k .
- The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces equals n , and this happens if and only if (i) the characteristic polynomial factors completely into linear factors and (ii) the dimension of the eigenspace for each λ_k equals the multiplicity of λ_k .
- If A is diagonalizable and $\mathcal{B}_k$ is a basis for the eigenspace corresponding to λ_k for each k , then the total collection of vectors in the sets $\mathcal{B}_1, \dots, \mathcal{B}_p$ forms an eigenvector basis for $\mathbb{R}^n$.

EXAMPLE 6 Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 1 & 4 & -3 & 0 \\ -1 & -2 & 0 & -3 \end{bmatrix}$$

SOLUTION Since A is a triangular matrix, the eigenvalues are 5 and -3 , each with multiplicity 2. Using the method in Section 5.1, we find a basis for each eigenspace.

Basis for $\lambda = 5$: $\mathbf{v}_1 = \begin{bmatrix} -8 \\ 4 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -16 \\ 4 \\ 0 \\ 1 \end{bmatrix}$

Basis for $\lambda = -3$: $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{v}_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

The set $\{\mathbf{v}_1, \dots, \mathbf{v}_4\}$ is linearly independent, by Theorem 7. So the matrix $P = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_4]$ is invertible, and $A = PDP^{-1}$, where

$$P = \begin{bmatrix} -8 & -16 & 0 & 0 \\ 4 & 4 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

```
>> M = [A - 5*eye(4) zeros(4,1)]
```

```
M =
```

```

0      0      0      0      0
0      0      0      0      0
1      4     -8      0      0
-1     -2      0     -8      0
```

```
>> rref(M)
```

```
ans =
```

```

1      0      8     16      0
0      1     -4     -4      0
0      0      0      0      0
0      0      0      0      0
```

```
>> M = [A - (-3)*eye(4) zeros(4,1)]
```

```
M =
```

```

8      0      0      0      0
0      8      0      0      0
1      4      0      0      0
-1     -2      0      0      0
```

```
>> rref(M)
```

```
ans =
```

```

1      0      0      0      0
0      1      0      0      0
0      0      0      0      0
0      0      0      0      0
```